

Methods To Implement the Nested Json APIs

==> Method 1:

--> This is the products API data we want to show on our UI

```
{
  "products": [
    {
      "id": 1,
      "title": "Essence Mascara Lash Princess",
      "description": "The Essence Mascara Lash Princess is a popular mascara known for its volumizing and lengthening effects. Achieve dramatic lashes with this long-lasting and cruelty-free formula.",
      "category": "beauty",
      "price": 9.99,
      "discountPercentage": 10.48,
      "rating": 2.56,
      "stock": 99,
      "tags": [
        "beauty",
        "mascara"
      ],
      "brand": "Essence",
      "sku": "BEA-ESS-ESS-001",
      "weight": 4,
      "dimensions": {
```

```
    "width": 15.14,
    "height": 13.08,
    "depth": 22.99
  },
  "warrantyInformation": "1 week warranty",
  "shippingInformation": "Ships in 3-5 business
days",
  "availabilityStatus": "In Stock",
  "reviews": [
    {
      "rating": 3,
      "comment": "Would not recommend!",
      "date": "2025-04-30T09:41:02.053Z",
      "reviewerName": "Eleanor Collins",
      "reviewerEmail":
"eleanor.collins@x.dummyjson.com"
    },
    {
      "rating": 4,
      "comment": "Very satisfied!",
      "date": "2025-04-30T09:41:02.053Z",
      "reviewerName": "Lucas Gordon",
      "reviewerEmail":
"lucas.gordon@x.dummyjson.com"
    },
    {
      "rating": 5,
      "comment": "Highly impressed!",
      "date": "2025-04-30T09:41:02.053Z",
```

```

        "reviewerName": "Eleanor Collins",
        "reviewerEmail":
"eleanor.collins@x.dummyjson.com"
    }
],
"returnPolicy": "No return policy",
"minimumOrderQuantity": 48,
"meta": {
    "createdAt": "2025-04-30T09:41:02.053Z",
    "updatedAt": "2025-04-30T09:41:02.053Z",
    "barcode": "5784719087687",
    "qrCode":
"https://cdn.dummyjson.com/public/qr-code.png"
},
"images": [
    "https://cdn.dummyjson.com/product-
images/beauty/essence-mascara-lash-princess/1.webp"
],
"thumbnail":
"https://cdn.dummyjson.com/product-
images/beauty/essence-mascara-lash-
princess/thumbnail.webp"
},],}

```

**--> This is the Model Class of Products for understanding better
Decoding of Json Data**

// product_model.dart

```
class Product {  
    final int id;  
    final String title;  
    final String description;  
    final double price;  
    final double rating;  
    final List<String> images;  
    final double height;  
    final List<Review> reviews;  
  
    Product({  
        required this.id,  
        required this.title,  
        required this.description,  
        required this.price,  
        required this.rating,  
        required this.images,  
        required this.height,  
        required this.reviews,  
    });
```

```
factory Product.fromJson(Map<String, dynamic> json) {  
  return Product(  
    id: json['id'],  
    title: json['title'],  
    description: json['description'],  
    price: json['price'].toDouble(),  
    rating: json['rating'].toDouble(),  
    images: List<String>.from(json['images']),  
    height: json['dimensions']['height'].toDouble(),  
    reviews: List<Review>.from(json['reviews'].map((x) =>  
Review.fromJson(x))),  
  );  
}  
}
```

```
class Review {  
  final int rating;  
  final String comment;  
  final String reviewerName;  
  
  Review({
```

```
required this.rating,  
required this.comment,  
required this.reviewerName,  
});
```

```
factory Review.fromJson(Map<String, dynamic> json) {  
  return Review(  
    rating: json['rating'],  
    comment: json['comment'],  
    reviewerName: json['reviewerName'],  
  );  
}  
}
```

==> Access Data in UI Flutter

```
// UI Example  
Center(  
  child: Container(  
    height: 400,
```

```
margin: EdgeInsets.all(16),
padding: EdgeInsets.all(20),
decoration: BoxDecoration(
  gradient: LinearGradient(
    colors: [Colors.purpleAccent, Colors.deepPurpleAccent],
    begin: Alignment.topLeft,
    end: Alignment.bottomRight,
  ),
  borderRadius: BorderRadius.circular(20),
  boxShadow: [
    BoxShadow(
      color: Colors.black26,
      blurRadius: 10,
      spreadRadius: 2,
    ),
  ],
),
child: Column(
  crossAxisAlignment: CrossAxisAlignment.start,
  children: [
    Text(
```

```
        product.title,
        style: TextStyle(fontSize: 22, fontWeight: FontWeight.bold,
color: Colors.white),
    ),
    SizedBox(height: 10),
    Image.network(product.images[0], height: 100),
    SizedBox(height: 10),
    Text(
        "Rating: ${product.rating.toString()}",
        style: TextStyle(color: Colors.white),
    ),
    Text(
        "Height: ${product.height.toString()} cm",
        style: TextStyle(color: Colors.white),
    ),
    SizedBox(height: 10),
    Text(
        "Latest Review:",
        style: TextStyle(fontWeight: FontWeight.bold, color:
Colors.white),
    ),
```



```

        Text(
            "${product.reviews[0].comment}" -
            "${product.reviews[0].reviewerName}',
            style: TextStyle(color: Colors.white70),
        ),
    ],
),
),
);

```

==> Parse Json Data

```

List<Product> productList = (jsonData['products'] as List)
    .map((product) => Product.fromJson(product))
    .toList();

```

// jsonData["products"] is the list starting at the top of our json data coming from the API

==> Method 2:

--> This is the User API data

```
[  
  {  
    "id": 1,  
    "name": "Leanne Graham",  
    "username": "Bret",  
    "email": "Sincere@april.biz",  
    "address": {  
      "street": "Kulas Light",  
      "suite": "Apt. 556",  
      "city": "Gwenborough",  
      "zipcode": "92998-3874",  
      "geo": {  
        "lat": "-37.3159",  
        "lng": "81.1496"  
      }  
    },  
    "phone": "1-770-736-8031 x56442",  
    "website": "hildegard.org",
```

```
"company": {  
  "name": "Romaguera-Crona",  
  "catchPhrase": "Multi-layered client-server neural-  
net",  
  "bs": "harness real-time e-markets"  
}  
},  
]
```

--> this is the .fromMap() method to decode the complex json data

==> .formMap() Method

```
class UserApi {  
  final String name;  
  final String email;  
  final String companyName;  
  final String address;
```

```
UserApi({  
    required this.name,  
    required this.email,  
    required this.companyName,  
    required this.address,  
});
```

```
factory UserApi.fromMap(Map<String, dynamic>  
map) {  
    final address = map['address'];  
    final company = map['company'];  
  
    return UserApi(  
        name: map['name'] ?? "",  
        email: map['email'] ?? "",  
        address: '${address['street']}, ${address['city']}',  
        companyName: company['name'] ?? "",  
    );
```

}

// after this we will be able to decode this complex json data into simple variables in the model class

==> Method 3:

--> this is the json API data about products of carts
{ "carts": [{ "id": 1, "products": [{ "id": 168, "title": "Charger SXT RWD", "price": 32999.99, "quantity": 3, "total": 98999.97, "discountPercentage": 13.39, "discountedTotal": 85743.87, "thumbnail": "https://cdn.dummyjson.com/products/images/vehicle/Charger%20SXT%20RWD/thumbnail.png" }, { "id": 78, "title": "Apple MacBook Pro 14 Inch Space Grey", "price": 1999.99, "quantity": 2, "total": 3999.98, "discountPercentage": 18.52, "discountedTotal": 3259.18, "thumbnail": "https://cdn.dummyjson.com/products/images/laptops/Apple%20MacBook%20Pro%2014%20Inch%20Space%20Grey/thumbnail.png" }, { "id": 183, "title": "Green Oval Earring", "price": 24.99, "quantity": 5, "total": 124.95, "discountPercentage": 6.28, "discountedTotal": 117.1, "thumbnail": "https://cdn.dummyjson.com/products/images/wo

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```
jewellery/Green%20Oval%20Earring/thumbnail.png"
}, { "id": 100, "title": "Apple AirPods", "price":
129.99, "quantity": 5, "total": 649.95,
"discountPercentage": 12.84, "discountedTotal":
566.5, "thumbnail":
"https://cdn.dummyjson.com/products/images/mobile-accessories/Apple%20Airpods/thumbnail.png" }
], "total": 103774.85, "discountedTotal": 89686.65,
"userId": 33, "totalProducts": 4, "totalQuantity": 15 }
],}
```

--> this is the product class inside the carts

```
// product.dart
```

```
class Product {
  final int id;
  final String title;
  final double price;
  final int quantity;

  Product({
    required this.id,
    required this.title,
    required this.price,
```

```
    required this.quantity,  
  });
```

```
  factory Product.fromJson(Map<String, dynamic>  
    json) {  
    return Product(  
      id: json['id'],  
      title: json['title'],  
      price: json['price'].toDouble(),  
      quantity: json['quantity'],  
    );  
  }  
}
```

--> this is the carts class outside the products list

```
// cart.dart  
import 'product.dart';  
  
class Cart {  
  final int id;  
  final List<Product> products;  
  
  Cart({  
    required this.id,  
    required this.products,  
  });
```

```

factory Cart.fromJson(Map<String, dynamic> json)
{
    var productsFromJson = json['products'] as List;
    List<Product> productList =
productsFromJson.map((productJson) =>
Product.fromJson(productJson)).toList();

    return Cart(
        id: json['id'],
        products: productList,
    );
}
}

```

--> this is the fetchData() from the API

```

// api_service.dart
import 'dart:convert';
import 'package:http/http.dart' as http;
import 'cart.dart';

Future<List<Cart>> fetchCarts() async {
    final response = await
http.get(Uri.parse('https://your-api-
url.com/carts'));

    if (response.statusCode == 200) {

```



```
final jsonData = jsonDecode(response.body);  
List cartsJson = jsonData['carts'];  
  
List<Cart> carts = cartsJson.map((cartJson) =>  
Cart.fromJson(cartJson)).toList();  
return carts;  
} else {  
throw Exception('Failed to load carts');  
}  
}
```